

MEC E 686 Assessment and Analysis of Biomechanical Motion

Winter 2026

Instructor: Dr. Hossein Rouhani

Assignment 2 (100 marks)

Important assignment information: This assignment is due on **Thursday, March 26, 2026**. Please submit a single PDF file of your assignment solution via the Canvas course webpage before 11:00 am on this date. In completing the assignment, **students must provide complete solutions:** the process used in answering the assignment questions should be described, and any MATLAB code developed as part of the solution process should be referenced and included in an appendix.

The following files have been included with this assignment:

MecE686_Assignment2_Guide.pptx: Contains the equations required to answer the assignment questions.

FPEMG_P_#_T_#.txt: Time series for the kinetic and electromyographic data recorded during the walking trials. The ground reaction force (components abbreviated as F_x , F_y , and F_z) and moment (abbreviated as M_x , M_y , and M_z) were recorded in units of N and N-mm, respectively. The center of pressure (abbreviated as C_x , C_y , and C_z) was recorded in units of mm and electromyograms were recorded in units of V. Each file contains the data for a single trial. For example, FPEMG_P_2_T_4.txt contains the data for the fourth trial of Participant 2. The 'Frame' column labels the motion capture samples. The 'Sub-Frame' column labels the kinetic and electromyogram samples recorded within each motion capture frame. The electromyogram abbreviations correspond to those defined in the Biomechanics Assignments Overview document. Note that the kinetic data (measured by force-plate) are expressed in the laboratory coordinate system.

Using the information contained in the files included with this assignment and the previous one as well as the Biomechanics Assignments Overview document, answer the following questions:

1. (5 marks) For each trial, obtain the kinetic data (ground reaction force (GRF) applied to the body, centre of pressure (COP), free frictional torque around the COP (T_z) applied to the body) and downsample them from 2,000 to 100 Hz (i.e., so that its sampling frequency matches that of the motion capture data) by extracting the kinetic data samples with Frames and Sub-Frames that match those of samples in the motion capture data included with Assignment 1 (i.e., you should extract rows 1, 21, 41, etc. in the kinetic data for each trial). **Note that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be**

referenced and included in an appendix.

2. **(5 marks)** For each trial, motion capture sample, and body side, calculate the center of rotation position (JCR) of the ground, ankle, knee, and hip joints in units of metres (*this should be done according to the equations provided in the Assignment 2 Guide*). **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
3. **(5 marks)** Using the results of Question 2, for each trial, motion capture sample, and body side, calculate the center of mass position (COM) of the foot, shank, and thigh segments in units of metres (*this should be done according to the equations provided in the Biomechanics Assignment 2 Guide*). **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
4. **(5 marks)** Using the results of Questions 1 and 3 as well as of Question 4 in Assignment 1, for each trial, extract the GRF, centre of pressure (COP), free frictional torque around the COP (T_z), and COM for the force plate gait cycle of the body side of the foot that landed on the force plate. For example, for the first trial of Participant 1, you should extract the GRF, COP, and T_z , and the COM of the right body side from samples 332 to 459 and discard the rest of the data for this trial (i.e., the remaining GRF, COP, and T_z data and right body side COM data as well as all the left body side COM data for this trial should be discarded). **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
5. **(5 marks)** **Note: For this question and the following ones, analyze only the data extracted in Question 4.** Using the results of Question 4, for each trial, filter the GRF, COP, T_z and COM using a fourth-order, zero phase-shift, low-pass Butterworth filter with a cut-off frequency of 10 Hz. This should be done using the MATLAB commands described in Question 9 of Assignment 1. **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
6. **(10 marks)** Using the results of Question 5, for each trial, calculate the angular velocity and acceleration of the foot, shank, and thigh in units of rad/s and rad/s² and the linear acceleration of their COM in units of m/s² (*this should be done according to the equations provided in the Biomechanics Assignment 2 Guide*). **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
7. **(15 marks)** Using the results of Questions 5 and 6, calculate the joint reaction forces, net joint moment and joint power for the ankle, knee, and hip joints in units of N, N.m, and W (*this should be done according to the equations provided in the Biomechanics Assignment 2 Guide*). The body masses of Participants 1, 2, and 3 were 76.5, 65.8, and 82.7 kg,

respectively. **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**

8. **(5 marks)** Using the results of Question 7, use linear interpolation to time normalize the joint reaction forces, net joint moment and joint power time series for each trial to 101 samples. This should be done using the MATLAB command described in Question 6 of Assignment 1. **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
9. **(5 marks)** Using the results of Question 8, for each participant, calculate the ensemble average (i.e., mean and standard deviation (SD)) joint reaction forces, net joint moment and joint power time-series in the laboratory coordinate system and then in the local coordinate system of the proximal segment. **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
10. **(15 marks)** Using the results of Question 9, for each participant, create
 - a 3-by-3 (one row for each force component and one column for each joint) figure that presents the ensemble average joint reaction forces expressed in the local coordinate system of the proximal segment in units of N.
 - a 3-by-3 (one row for each moment component and one column for each joint) figure that presents the ensemble average net joint moment expressed in the local coordinate system of the proximal segment in units of N.m.
 - a 1-by-3 (one column for each joint) figure that presents the ensemble average joint power in units of W.

The ensemble average joint forces should be presented as the mean ± 1 SD bounds. Additionally, for a given force, moment, and power component and joint, the ensemble average joint forces. Note that, since Participant 3 walked in the $-Y$ and $+Y$ directions of the laboratory coordinate system for left and right foot force plate steps, respectively (i.e., opposite to the directions walked by Participants 1 and 2), when plotted, the X and Y components of the joint forces of Participant 3 will be mirrored about the horizontal plot axis relative to those of Participants 1 and 2 (you will not need to adjust for this difference). **Provide the created figures (three figures total). Additionally, the MATLAB code used to create the figures should be referenced and included in an appendix.**